Pack current, voltage and load switch

File: sense.kicad_sch

MCP16331 5V / LDO 3.3V

File: power.kicad_sch

ESP32-C3 and user interfaces

File: controller.kicad_sch

PWR_FLAG

GND

PWR_FLAG

SYS_PLUS

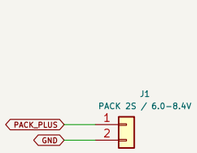
PWR_FLAG

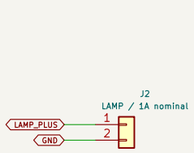
5V

Sheet: /
File: flashlight.kicad_sch

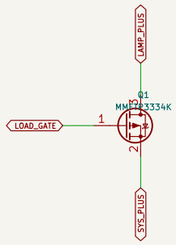
Title: 2S flashlight controller / bottom assembly

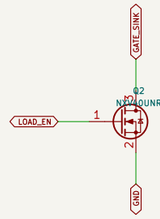
Size: A4	Date:	Rev: 0.2
KiCad E.D.A. 10.0.5		Id: 1/4

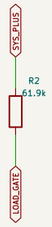


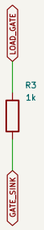


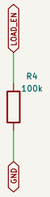


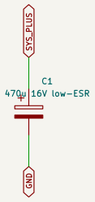


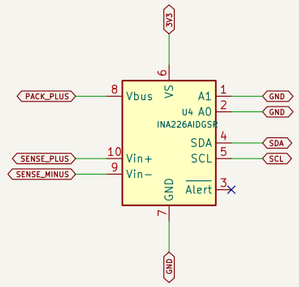


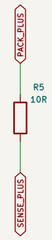


















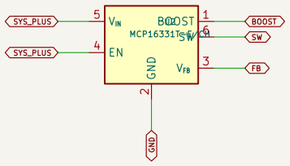
R1 = 47 mOhm (reference WSL2512R0470FEA); INA226 shunt range +/−1.743 A. Update firmware calibration.
MMFTP3334K + NXV40UNR: slow lamp switching. Not validated for high−frequency PWM.

2S 6.0–8.4V; bottom assembly; verify sensor and mechanical dimensions

Sheet: /Pack current, voltage and load switch/
File: sense.kicad_sch

Title: Pack measurement and lamp switch

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.5		Id: 2/4









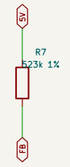


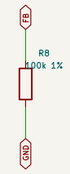


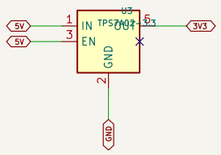
















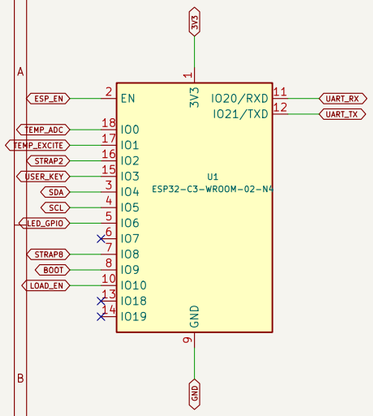
R7/R8 = 523k/100k: 4.984 V nominal, 8 uA divider. Verify ripple/start-up/load steps on prototype.
MCP16331 has no specified maximum feedback resistance or guaranteed maximum FB bias current.

2S 6.0–8.4V; bottom assembly; verify sensor and mechanical dimensions

Sheet: /MCP16331 5V / LDO 3.3V/
File: power.kicad_sch

Title: 5V buck and 3.3V regulator

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.5		Id: 3/4

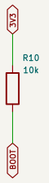






















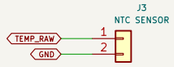








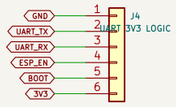












PROVISIONAL: NTC 10k/B3950, 0..100 C. GPIO1 excitation; wait 15 ms, sample ADC1_CH0, drive GPIO1 low.
ADC ATTEN3 / 12 dB: calibrated range 0..2.5 V; 100 nF at GPIO0. Confirm actual sensor before assembly.

2S 6.0–8.4V; bottom assembly; verify sensor and mechanical dimensions

Sheet: /ESP32-C3 and user interfaces/
File: controller.kicad_sch

Title: ESP32-C3, buttons, LED and temperature

Size: A3	Date:	**Rev: 0.2**
KiCad E.D.A. 10.0.5		Id: 4/4